universally understood representations that are, in many cases, already integrated with existing workflows.

MAPS

As we saw in Chapter 1, the 2D geometric representation of a physical area that is provided by a geographical map forms the basis of many digital twins. Maps are spatial reference frameworks that provide geographic context, location relationships and environmental understanding that many people are familiar with, having been taught to use them from a young age.

Soil maps, geological surveys, and environmental hazard zones documented in specialized maps provide critical information for understanding foundation conditions, seismic risks, and environmental constraints that affect system design and operation. This contextual information is essential for creating digital twins that accurately represent how physical systems interact with their environmental setting.

Geographic Information Systems (GIS) serve as the authoritative foundation for spatially-referenced digital twins, providing standardized frameworks for capturing, storing, analyzing, and visualizing geospatial data with precise coordinate reference systems and established accuracy standards.

FLOORPLANS

Floorplans instantly convey room count, size, and layout using simple, scaled diagrams and standard architectural symbols. While 3D models contain the same information, floorplans deliver it more efficiently, providing a clear, uncluttered view of spatial logic.

These precise measurements enable floorplans to serve as powerful computational tools. Digital formats allow path-finding algorithms (like A* and Dijkstra’s) to calculate optimal routes between any two points. This is crucial for comprehensive digital twins that simulate movement, such as tracking people in an office or robots in a warehouse.